Subject: Mathematics
Class: Class X (CBSE)

CBSE CLASS 10 – WEEKLY TEST 01
Chapter 1: Real Numbers

Time: 45 Minutes
MM: 20

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided in question subparts.
2. **Section A** comprises 4 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 1 Very Short Answer (VSA) question of 2 marks.
4. **Section C** comprises 3 Short Answer (SA) questions of 3 marks each.
5. **Section D** comprises 1 Case Study based question of 4 marks with internal subparts.
6. Use of calculators is strictly prohibited. All mathematical steps must be clearly shown.

SECTION A: Multiple Choice & Assertion-Reason (5 × 1 = 5 Marks)

Q1. The HCF of the smallest prime number and the smallest composite number is: [1]
(a) 1 (b) 2 (c) 4 (d) 8

Q2. The LCM of the smallest two-digit composite number and the smallest composite number is: [1]
(a) 12 (b) 4 (c) 20 (d) 40

Q3. If $a = 2^3 \times 3$, $b = 2 \times 3 \times 5$, $c = 3^n \times 5$ and $\text{LCM}(a, b, c) = 2^3 \times 3^2 \times 5$, then the value of n is: [1]
(a) 1 (b) 2 (c) 3 (d) 4

Q4. If two positive integers a and b are written as $a = x^3y^2$ and $b = xy^3$, where x, y are prime numbers, then $\text{HCF}(a, b)$ is: [1]
(a) xy (b) xy^2 (c) x^3y^3 (d) x^2y^2

Q5. **Assertion (A):** The HCF of 11 and 17 is 1.
Reason (R): If p and q are distinct prime numbers, then their highest common factor is always 1. [1]
(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Type (1 × 2 = 2 Marks)

Q6. Explain why $7 \times 11 \times 13 + 13$ and $7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 + 5$ are composite numbers. [2]

SECTION C: Short Answer Type (3 × 3 = 9 Marks)

Q7. Prove that $\sqrt{2}$ is an irrational number. [3]

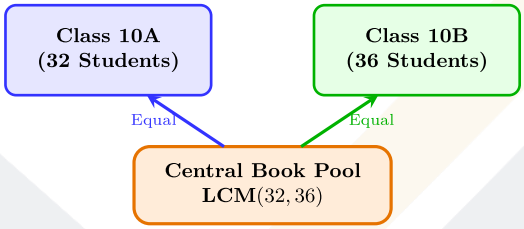
Q8. Prove that $\sqrt{3}$ is an irrational number. [3]

Q9. Prove that $\sqrt{5}$ is an irrational number. [3]

SECTION D: Case Study Based Question (1 × 4 = 4 Marks)

Q10. CASE STUDY: CASE STUDY 01: SCHOOL READING CAMPAIGN LIBRARY BOOK DISTRIBUTION [4 Marks]

To foster independent reading habits among Class 10 students, a school administration establishes a dedicated classroom library. Class 10 has two sections: Section A with 32 students and Section B with 36 students. The head librarian orders books from a central warehouse and wants to acquire a minimum quantity of books such that they can be distributed completely and equally among the students of Section A or Section B without any books remaining.



(i) Express the student strengths 32 and 36 as products of their prime factors. [1]

(ii) Determine the minimum number of books the librarian must acquire to fulfill the condition. [2]

(iii) If 720 books are acquired instead, how many books does each student of Section A and Section B receive respectively? [1]

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